

ARARAT, Australia

WMO: 948340

Lat: 37.277S

Lon: 142.981E

Elev: 296

StdP: 97.82

Time Zone: 10.00 (AUS)

Period: 94-05

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	-1.2	0.4	-3.5	2.9	9.5	-1.8	3.4	8.9	13.4	9.0	13.0	9.6	0.6	200	0.556

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	13.9	34.0	17.5	31.8	17.1	29.7	16.5	19.9	27.7	18.9	27.7	18.0	25.9	4.9	340

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
17.8	13.3	22.0	16.4	12.1	20.7	15.0	11.1	20.0	58.3	27.9	54.8	27.7	51.8	26.0	23.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
14.8	13.0	10.4	-3.6	38.9	0.9	1.7	-4.3	40.1	-4.9	41.0	-5.4	42.0	-6.1	43.2		
			-4.1	21.7	1.0	1.1	-4.8	22.5	-5.3	23.1	-5.9	23.7	-6.6	24.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	12.9	19.0	19.6	16.5	12.8	10.3	8.5	8.0	8.4	10.1	11.5	14.2	16.6
	DBStd	5.09	4.05	4.09	3.55	3.09	2.47	2.27	1.86	2.27	2.89	2.95	3.59	4.13
	HDD10.0	263	0	0	0	7	26	54	66	61	32	14	3	0
	HDD18.3	2174	40	31	81	169	249	294	321	309	248	212	135	86
	CDD10.0	1328	280	268	202	92	35	10	3	11	34	62	128	204
	CDD18.3	196	62	65	23	3	0	0	0	0	1	1	9	32
	CDH23.3	2933	893	856	334	37	1	0	0	0	2	37	175	598
	CDH26.7	1241	409	388	111	4	0	0	0	0	8	51	270	
Wind	WSAvg	4.0	5.0	4.5	3.7	3.3	2.8	3.5	3.3	3.6	4.1	4.2	4.7	4.9
Precipitation	PrecAvg	594	37	34	32	43	54	58	63	70	64	62	44	35
	PrecMax	909	137	226	101	110	144	128	114	135	117	202	76	102
	PrecMin	302	2	0	5	4	7	10	17	12	18	13	11	0
	PrecStd	121	27	44	24	32	31	31	27	29	30	39	18	29
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	37.3	37.1	32.2	26.6	21.7	16.3	15.0	18.8	23.0	27.7	31.4	35.8
		MCWB	18.9	19.1	17.3	14.2	12.6	11.9	10.4	12.9	14.2	14.3	15.8	17.0
	2%	DB	33.8	33.9	29.5	24.2	19.2	14.5	13.3	15.9	19.7	23.4	28.6	32.6
		MCWB	17.7	18.0	16.5	14.2	11.9	11.5	9.4	10.4	13.1	13.6	15.2	16.1
	5%	DB	31.2	31.3	27.1	21.9	16.8	13.2	12.1	14.2	17.6	20.7	25.3	29.9
		MCWB	17.4	17.7	15.8	13.5	11.6	10.4	9.0	9.9	12.3	12.8	14.6	15.6
	10%	DB	28.6	28.7	24.6	19.7	15.0	12.1	11.2	12.7	15.6	18.1	22.5	26.7
		MCWB	16.6	17.1	15.2	13.1	11.2	9.8	8.6	9.2	11.3	12.2	13.7	15.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	20.8	22.0	19.7	16.2	14.5	13.9	11.2	14.2	16.1	16.1	18.7	19.4
		MCDB	28.7	30.5	27.9	20.9	17.5	14.6	13.2	17.5	20.4	22.3	25.0	25.4
	2%	WB	19.4	20.6	18.0	15.2	13.3	11.9	10.2	11.6	14.3	14.7	17.1	18.1
		MCDB	30.3	26.6	25.5	21.2	16.3	13.7	12.0	14.2	18.6	20.0	24.1	25.4
	5%	WB	18.4	19.4	16.6	14.4	12.4	10.9	9.6	10.5	12.8	13.7	15.8	16.9
		MCDB	27.8	26.9	23.8	20.4	15.6	12.8	11.3	13.4	16.2	19.3	22.8	25.9
	10%	WB	17.5	18.4	15.7	13.4	11.7	10.0	9.1	9.6	11.6	12.6	14.5	15.9
		MCDB	25.9	26.7	23.4	18.8	14.8	11.6	10.8	12.2	15.0	17.8	20.7	25.2
Mean Daily Temperature Range	5% DB	MDBR	14.3	13.9	12.6	11.5	9.2	6.7	6.5	7.7	9.0	10.6	12.2	14.1
		MCDBR	20.0	19.0	17.9	16.0	13.4	7.9	7.7	10.4	12.0	16.4	18.4	21.4
		MCWBR	6.5	5.9	6.8	7.4	7.5	5.2	4.9	6.2	6.9	8.2	7.5	7.0
	5% WB	MCDBR	15.8	13.8	13.9	13.0	10.8	5.9	6.8	9.3	10.5	13.4	13.5	15.8
		MCWBR	6.1	5.5	6.0	6.9	6.6	4.3	4.7	6.1	6.8	7.3	6.8	6.2
Clear-Sky Solar Irradiance	taub	0.341	0.331	0.316	0.307	0.297	0.288	0.287	0.300	0.325	0.329	0.333	0.328	
	taud	2.473	2.520	2.554	2.559	2.564	2.583	2.582	2.526	2.441	2.445	2.470	2.506	
	Ebn at Noon	993	980	956	904	853	834	857	897	929	969	994	1010	
	Edh at Noon	116	106	95	82	71	65	69	83	104	113	116	113	
All-Sky Solar Radiation	RadAvg	7.46	6.54	5.16	3.60	2.41	1.91	2.08	2.95	4.11	5.47	6.47	7.26	
	RadStd	0.43	0.32	0.27	0.22	0.12	0.15	0.13	0.19	0.28	0.45	0.43	0.40	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (36 neighbors)	+0.48	N/A	N/A	+0.88	+0.41	N/A	-13	-127	+169	+54

Nomenclature: See separate page